Sime Darby Property

SIME DARBY PROPERTY PRESENTS · TOWNSHIP OF TOMORROW

GreenLoop

A smart recycling ecosystem for the township of tomorrow, recycling that pays for itself.

♻️ **Scan · Earn · Redeem, one self-funding loop**

"Your ideas could help shape how people live tomorrow."



GreenLoop

Smart Recycling Ecosystem · recycling that pays for itself

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🔗 **Live demo: greenloop-tan.vercel.app**

💻 **Code: github.com/NajibAzir/greenloop**

Malaysia wants to recycle more, but it still ends up in landfill

~38,000 t/day

solid waste generated daily, about **82%** still goes to landfill

~30%

national recycling rate, below the **40%-by-2025** target

~40%

of landfilled waste is actually **recyclable**, value lost

In a township, it stalls for three reasons:

No incentive

Effort, no reward, so resident turnout stays low.

Seen as a cost

Recycling is a cost centre, so programmes get cut.

Blind collection

Trucks run fixed routes to half-empty bins, wasting fuel.

Sources: UNIDO (2025) · Asia News Network (2026) · KPKT Circular Economy Blueprint 2025-2035. Figures indicative.

Three groups, one loop

Residents

WHAT THEY WANT

Households that want convenient recycling and an everyday reward for taking part.

Mall Tenants

WHAT THEY WANT

F&B / retail (e.g. Elmina Lakeside Mall) wanting to offset monthly service charges from packaging waste.

Developer Ops

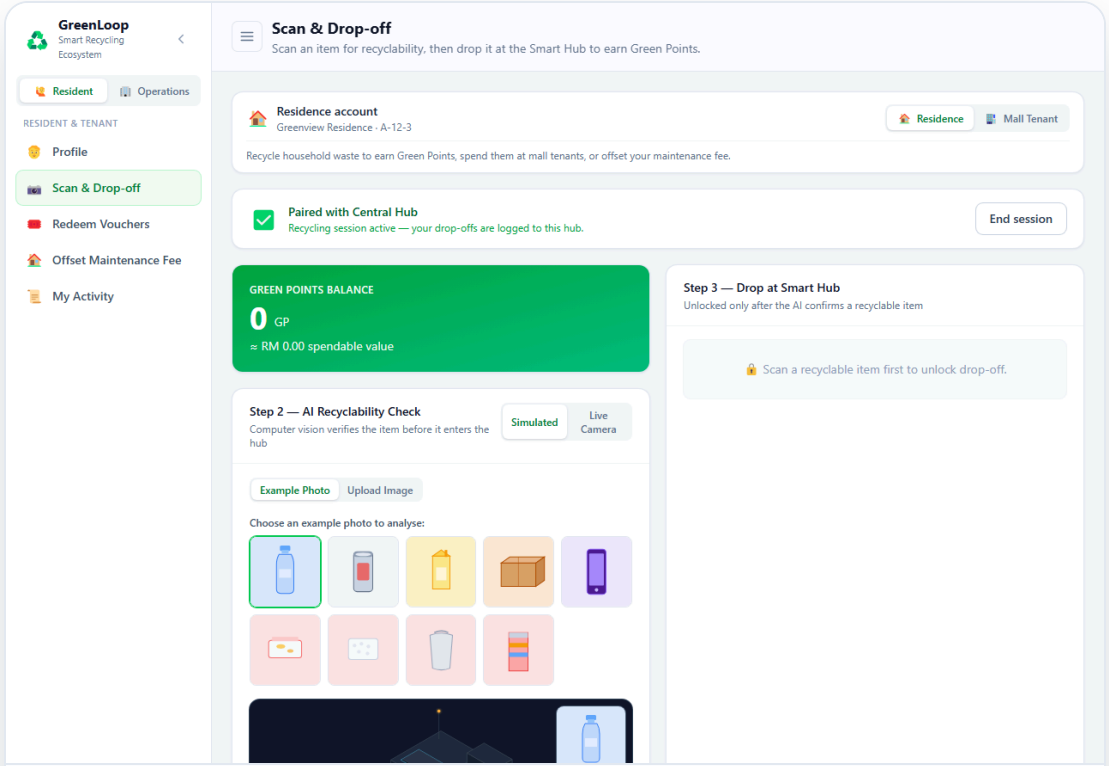
WHAT THEY WANT

Sime Darby Property wants visibility, lower haulage cost and measurable ESG outcomes.

📍 **Grounded in real locations, City of Elmina (Elmina West / East, Denai Alam, Bukit Subang), Elmina Lakeside Mall, Bandar Bukit Raja.**

GreenLoop closes the loop: recycling becomes a self-funding economy

- **Scan & earn**, AI verifies the item; residents earn Green Points (GP).
- **Spend like cash**, redeem GP at mall tenants or offset monthly fees.
- **Funded by savings**, avoided haulage + material sales + carbon value pay for it.
- **Collect smart**, AI dispatches trucks only to full bins.



Resident Scan & Drop-off, live prototype

A funded, verified, intelligent loop, not just another points app

✓ GREENLOOP

- ✓ Rewards **funded** by avoided haulage + material sales + carbon value
- ✓ **AI vision verifies** each item before it is accepted
- ✓ **Live AI dispatch**, trucks sent only to full bins, on real road routing
- ✓ Residents **and** mall tenants served by one engine

VS

✕ TYPICAL APP

- ✕ Rewards paid from a subsidy that runs out
- ✕ "Trust me, it's recyclable", no verification
- ✕ Fixed routes to half-empty bins
- ✕ Built for a single user type

Residents › Hubs › Operators › Rewards: one closed loop

1 → 2 → 3 → 4 → 5

Scan
Resident scans an item; AI vision confirms it's recyclable.

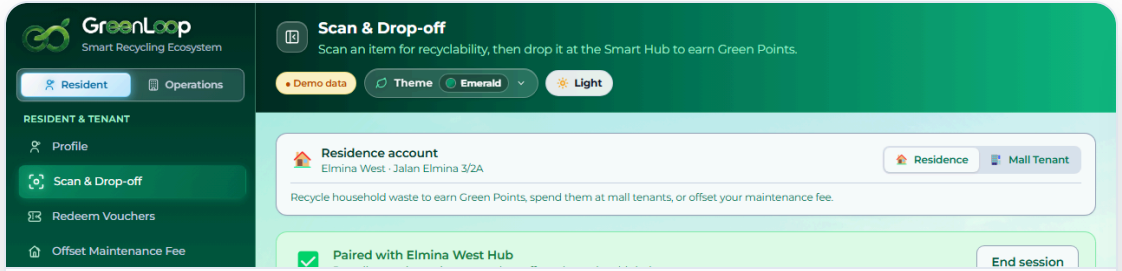
Drop at Hub
Pair via hub QR; item accepted, GP credited instantly.

AI Dispatch
At 80% fill, a truck is scheduled to full bins only.

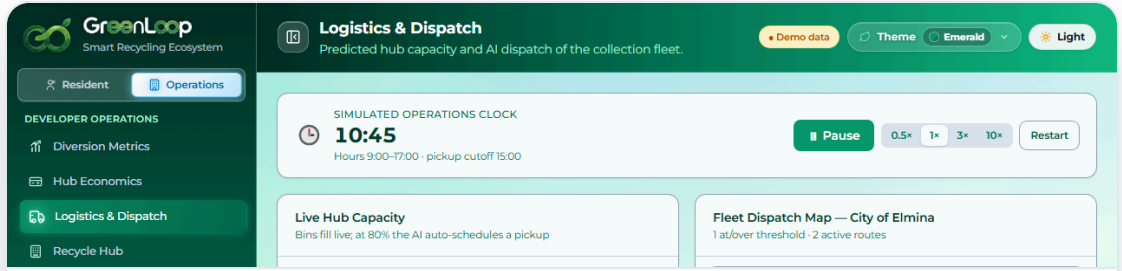
Recover & Sell
Material is diverted from landfill and sold to recyclers.

Value Funds Loop
Savings + sales + carbon fund GP & fee offsets.

Savings + material sales + carbon value fund the Green Points and fee offsets, and the loop repeats.



Resident, Smart Hub drop-off



Operator, AI Logistics & Dispatch

Measurable outcomes, honest about demonstrated vs projected

Environmental

Tonnes diverted from landfill; CO_2e avoided; directly supports the national 40% recycling target.

Community

Residents earn GP; GP offsets maintenance fees; higher, sustained participation over time.

Operational

Lower haulage cost, live hub & fleet visibility, a self-funding programme, not a subsidy.

✓ DEMONSTRATED

Working AI scan, GP economy, live dispatch, 6-hub map.

~ MODELLED

~10.9 t/month diverted and an operating surplus.

! NEEDS VALIDATION

Bin hardware, real haulage cost, recycler & carbon prices.

We show what's real today and label what still needs proving

The hub pays for itself

MONTHLY ECONOMICS (INDICATIVE)	RM
+ Material sales (cash)	**19,662**
+ Avoided haulage	**2,508**
+ Carbon / ESG value	**1,767**
Total benefit	**23,936**
− Operating cost	**4,200**
− Green Points redeemed	**4,931**
Total cost	**9,131**

SIMPLE PAYBACK

5.7 mo

RM 85,000 setup ÷ RM 14,805/mo net

≈ RM 14,805 net / month

Conservative case (no carbon): net ≈ RM 13,000/mo, payback ~6.5 months, still self-funding.

⚠ **RM 230/t avoided-haulage is an assumption pending validation with Sime Darby's waste contractor.**

Reconciles with the live Hub Economics dashboard

Start with one hub at Elmina West, prove it, then scale

Month 1

Install & integrate

Smart Hub + bin hardware; link rewards and resident sign-up.

Month 2-3

Run & measure

Track diversion, participation and real cost vs the model.

Month 4

Review, go / no-go

Compare actuals to assumptions; confirm payback.

Month 5+

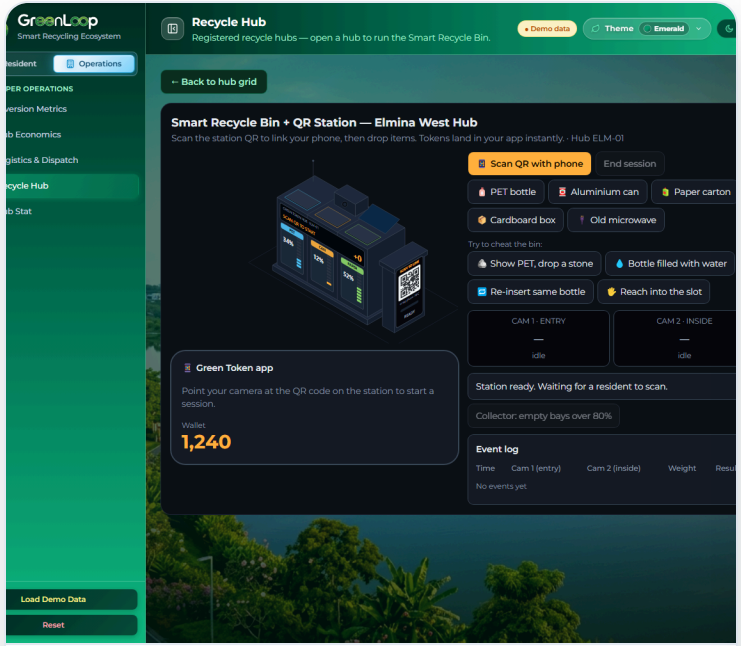
Scale

Roll out to Elmina East, Denai Alam, Elmina Lakeside Mall.

Stakeholders: SDP ops, waste contractor, mall tenants, residents, recyclers · Validate first: bin hardware, real RM/t haulage, recycler prices, POS, PDPA

One working ecosystem

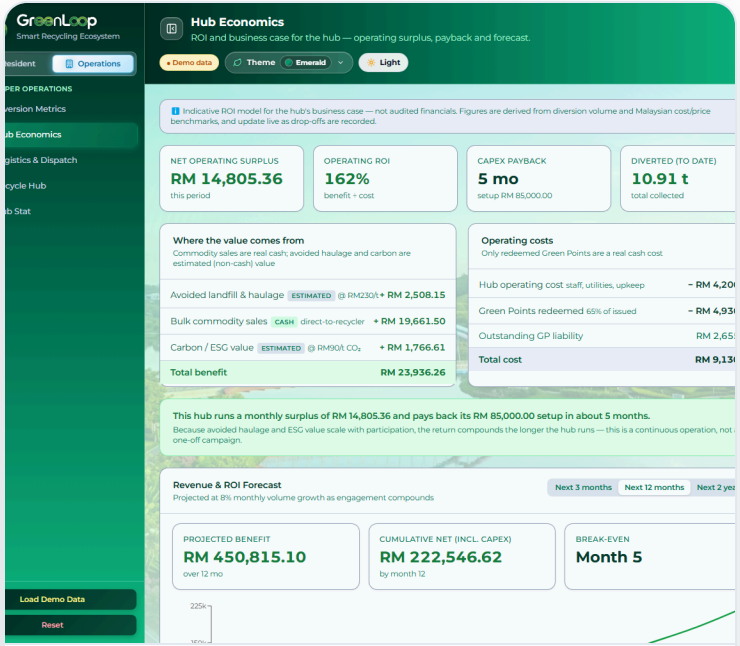
Resident app, smart bin, and the operator's command centre, all live in the prototype.



Smart Recycle Bin + QR station



Diversion metrics, actual vs projected



Hub economics, business case

Why GreenLoop

1. GreenLoop connects **AI-assisted recycling, resident rewards and smarter collection** into one township ecosystem, tackling both participation and the cost of collection, not just one.

2. We built a **deployed, interactive prototype** and a one-hub pilot model for Elmina West, and we're transparent about what is demonstrated vs what still needs real-world validation.

Our ask: a one-hub pilot at Elmina West plus access to operational data, so we can measure actual diversion, collection savings and payback before scaling.

Prove it at one hub. Then scale precinct by precinct.

Sime Darby Property

GreenLoop

Thank You

Recycling that pays for itself, for the township of tomorrow.

🔗 greenloop-tan.vercel.app

💻 github.com/NajibAzir/greenloop

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